SMART CABLE HEALTH PREDICTIVE SYSTEM

A PROJECT REPORT

Of

ECB1303 – EMBEDDED SYSTEM AND IOT DESIGN

Submitted by

K. SHAKTHI	**(8115U24EC101)**
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S.VARSHINI	**(8115U24EC119)**
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in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING

in

ELECTRONICS AND COMMUNICATION ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY
CHENNAI 600 025

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Under the Guidance of

Dr.R.SAMSON DANIEL

Associate Professor
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K. RAMAKRISHNAN COLLEGE OF ENGINEERING
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Under
ANNA UNIVERSITY, CHENNAI

NOVEMBER 2026

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BONAFIDE CERTIFICATE

Certified that this project report titled "**SMART CABLE HEALTH PREDICTIVE SYSTEM**" is the bonafide work of **SHAKTHI K (8115U24EC101), SHREE VIDHARSHANA S (8115U24EC104), VARSHINI S (8115U24EC119), YUKASHINI B (8115U24EC123)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported herein does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION BY THE CANDIDATES

We declare that to the best of our knowledge the work reported herein has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

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To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

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M1. To equip the students with latest technical, analytical and practical knowledge.

M2. To provide vibrant academic environment and Innovative Research culture.

M3. To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

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PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and the World: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

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PO11: Lifelong learning: Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

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We thank **Dr. M. MAHESWARI, M.Tech., Ph.D.**, Head of the Department of **ELECTRONICS AND COMMUNICATION ENGINEERING**, for providing her encouragement in pursuing this project.

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ABSTRACT

The Smart Cable Health predictive system is designed to continuously monitor the condition of electrical cables and prevent potential hazards caused by abnormal operating conditions. The proposed system uses an ESP32 microcontroller integrated with various sensors to detect changes in voltage, current-related magnetic field, temperature, and vibration around the cable.An AC voltage sensor monitors the voltage level of the 230V AC supply, while a Hall-effect sensor detects magnetic field variations associated with current flow. A temperature and humidity sensor (DHT11) monitors the surrounding environmental conditions, and a vibration sensor identifies unusual mechanical vibrations near the cable. The sensor readings are processed by the ESP32 to identify abnormal conditions based on predefined threshold values.

When any abnormal condition, such as overvoltage, unusual magnetic field variation, excessive temperature, or abnormal vibration, is detected, the system activates a buzzer and automatically disconnects the 230V AC supply to the monitored cable through a relay module. Additionally, the ESP32 uses Wi-Fi connectivity to transmit real-time monitoring data and fault notifications to the Blynk IoT cloud platform, allowing users to monitor cable conditions through a mobile dashboard. Thus, the proposed system provides continuous cable condition monitoring, automatic fault-based power disconnection, and remote monitoring capabilities. It aims to improve electrical safety, reduce the risk of cable-related failures, and support preventive maintenance in electrical installations.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	**ix**
1	**INTRODUCTION**	**1**
	1.1 INTRODUCTION	1
	1.2 EXISTING SYSTEM	2
	1.3 DRAWBACKS OF EXISTING SYSTEM	3
	1.4 NEED FOR PROPOSED WORK	4
	1.5 OBJECTIVES	4
	1.6 APPLICATIONS OF PROPOSED SYSTEM	5
2	**LITERATURE SURVEY**	**7**
3	**PROJECT DESCRIPTION**	**14**
	3.1 PROPOSED SYSTEM DESCRIPTION	14
	3.2 OVERALL BLOCK DIAGRAM AND EXPLANATION	15
	3.3 CIRCUIT DIAGRAM AND EXPLANATION	16
	3.4 POWER SUPPLY DESIGN	18
	3.5 HARDWARE COMPONENTS USED	19
	3.6 SOFTWARE USED	19
	3.7 SENSOR WORKING PRINCIPLES	21
	3.7.1 ZMPT101B AC VOLTAGE SENSOR	21
	3.7.2 KY-024 HALL EFFECT SENSOR	21
	3.7.3 SW-420 VIBRATION SENSOR	22
	3.7.4 DHT11 TEMPERATURE AND HUMIDITY SENSOR	22
	3.7.5 BUZZER	22
	3.8 RELAY-BASED AUTOMATIC POWER DISCONNECTION	23
	3.9 BLYNK IOT IMPLEMENTATION	26
	3.10 COMPLETE WORKING PRINCIPLE	27
4	**RESULTS AND DISCUSSION**	**29**
	4.1 HARDWARE PROTOTYPE AND EXPLANATION	29
	4.8 BLYNK DASHBOARD RESULTS	30
	4.9 OVERALL SYSTEM PERFORMANCE	31
5	**CONCLUSION AND FUTURE SCOPE**	**33**
	REFERENCES	**36**

LIST OF FIGURES

FIGURE NO	FIGURE NAME	PAGE NO
3.1	Overall block diagram of the Smart Cable Health Monitoring and Automatic Power Disconnection System	15
3.2	Connection diagram of the ESP32 with the sensors, buzzer and relay module	16
3.3	Power supply chain of the monitoring circuit	18
3.5	Flowchart of the intended operation of the system (relay branch is the design intention)	27
4.1	Hardware prototype of the Smart Cable Health Monitoring and Automatic Power Disconnection System	29
4.2	Placement of the sensors near the monitored cable	20
4.8	Blynk dashboard showing normal operation	30
4.9	Blynk dashboard showing fault status	31

LIST OF TABLES

TABLE NO	TABLE TITLE	PAGE NO
3.1	Monitored parameters and fault levels	15
3.3	ESP32 pin configuration	17
3.4	Hardware components used in the project	23
3.6	Difference between the monitoring circuit and the relay switching circuit	23
3.7	Blynk virtual pin (datastream) assignment	26
3.8	Intended system response for each condition	28
4.5	Status of the functions of the prototype	28

CHAPTER-1
INTRODUCTION

1.1 INTRODUCTION

Electrical energy is delivered to homes, laboratories, workshops and industries through cables, and the safe operation of every electrical load depends on the condition of the cable that feeds it. A cable is exposed to electrical stress from supply voltage variations, thermal stress from load current and ambient temperature, mechanical stress from vibration and handling, and electromagnetic effects from the current it carries. Over a period of time these stresses can damage the conductor and its insulation and may finally lead to short circuits, overheating, electric shock or fire. In many installations the condition of a cable is not observed at all until a visible failure occurs.

The rapid growth of low-cost microcontrollers, sensors and wireless connectivity has made it practical to observe the condition of electrical equipment continuously. The Internet of Things (IoT) allows sensor data to be collected by an embedded controller, transmitted over Wi-Fi to a cloud platform, and displayed on a mobile phone so that the condition of the equipment can be viewed from any location. When such monitoring is combined with an automatic protective action, the system can move from simple observation to active protection of the cable and of the equipment connected to it.

The present project, Smart Cable Health Monitoring and Automatic Power Disconnection System Using ESP32 and IoT, monitors a 230 V AC cable using four sensing elements. A ZMPT101B AC voltage sensor observes the supply voltage, a DHT11 sensor observes the temperature and humidity near the cable, an SW-420 sensor detects abnormal vibration, and a KY-024 Hall effect sensor observes the magnetic field around the cable. All sensors are connected to an ESP32 development board, which evaluates the readings against defined limits, drives a

buzzer, and publishes the readings and the overall status to a Blynk IoT mobile dashboard through Wi-Fi.

A distinguishing feature of the system is the separation between the monitoring circuit and the monitored cable. The monitoring circuit, consisting of the ESP32, the sensors and the buzzer, is powered from its own low-voltage supply derived through a step-down transformer, a bridge rectifier, a filter capacitor and a 7805 regulator. The monitored cable is connected to the mains through a 12 V relay module that switches only the Phase wire. When a fault is detected, the relay is intended to disconnect the monitored cable, while the ESP32 and the sensors stay powered and continue to observe the cable. This allows the system to report the fault and to detect when the abnormal condition has clear.

1.2 EXISTING SYSTEM

In conventional low-voltage installations, cable protection is provided mainly by fuses, miniature circuit breakers (MCBs), residual current circuit breakers (RCCBs) and, in some cases, voltage protection relays. A fuse or an MCB opens the circuit when the current exceeds a rated value for a certain time, and an RCCB opens the circuit when it detects leakage current to earth. These devices are reliable and are mandatory in electrical installations, but they respond to current-based faults and have no knowledge of the surrounding condition of the cable.

Voltage protection is available in the form of under-voltage and over-voltage protection relays and in the form of surge protectors and voltage stabilizers. These devices protect a load from supply voltage variation but do not observe the temperature, vibration or magnetic field around the cable itself.

Industrial installations use condition monitoring systems, thermal imaging, partial discharge measurement and periodic insulation testing to assess the health of cables. These methods are effective but are generally expensive, need trained personnel and specialised instruments, and are carried out periodically instead of continuously. Several IoT-based prototypes reported in the literature (Chapter 2)

concentrate on locating faults in underground cables after the fault has occurred, typically using a resistor-network model of the cable and a microcontroller, and display the result on an LCD or web page.

1.3 DRAWBACKS OF EXISTING SYSTEM

Although the existing protection devices and monitoring methods are useful, several limitations can be observed

- **Reactive protection:** fuses, MCBs and RCCBs act after a current or leakage fault has occurred and do not give early indication of abnormal voltage, heating or vibration.
- **Single-parameter operation:** a conventional device usually observes only one quantity, such as current or voltage, and does not combine several parameters to judge the overall health of a cable.
- **No remote visibility:** the user must be physically present to learn that a protection device has operated or that a cable is operating under abnormal conditions.
- **Cost and complexity of industrial monitoring:** thermal imaging, partial discharge measurement and online condition monitoring equipment are expensive and are not suited to small installations, laboratories and educational prototypes.
- **Fault location focus:** many reported IoT prototypes locate a fault in a cable after it has occurred and do not provide continuous multi-parameter health monitoring with an automatic protective response.
- **Loss of monitoring after disconnection:** if the monitoring electronics are powered from the same supply that is disconnected, monitoring stops at the moment when the fault status needs to be observed and reported.

1.4 NEED FOR PROPOSED WORK

A cable that is operated under high voltage, low voltage, abnormal heating, vibration or an abnormal magnetic field is under stress even before it fails. If these

conditions can be detected early and the supply to the cable can be removed automatically, the risk of damage to the cable and to the connected load can be reduced. A system that does this using inexpensive and widely available components is useful for laboratories, workshops, small industries and educational demonstrations, and it also provides a practical platform for studying condition monitoring and predictive maintenance.

The proposed work is needed because it combines four sensing parameters, local alarm indication, remote monitoring through a mobile dashboard and an automatic disconnection arrangement in a single low-cost embedded system. The use of a separate supply for the monitoring circuit is also necessary so that the controller can continue to observe the cable after the Phase supply has been disconnected and can detect that the fault has cleared.

A further need is transparency about what has been achieved. The prototype is a learning and demonstration system; it is not a certified protection device and is not a substitute for MCBs, RCCBs and fuses.

1.5 OBJECTIVES

The objectives of the project are:

1. To monitor the voltage of a 230 V AC cable using the ZMPT101B sensor and to identify low-voltage (below 200 V) and high-voltage (above 260 V) conditions.
2. To monitor the temperature and humidity near the cable using the DHT11 sensor and to identify the temperature fault level of 35 °C and above.
3. To detect abnormal vibration of the cable using the SW-420 sensor.
4. To monitor magnetic field variation around the cable using the KY-024 Hall effect sensor and to identify the fault level of 3000 and above.
5. To activate a buzzer when any monitored parameter enters the fault range.

6. To disconnect only the Phase supply of the monitored cable through a relay, while keeping the ESP32 and the sensors powered, and to reconnect the supply automatically after the fault clears (intended operation).
7. To display the sensor readings and the overall status (NORMAL / MIDDLE LEVEL / FAULT) on a Blynk IoT mobile dashboard through Wi-Fi.
8. To power the monitoring circuit from a separate regulated low-voltage supply built from a step-down transformer, bridge rectifier, filter capacitor and 7805 regulator.
9. To document clearly the present limitations of the prototype and the safety precautions required for 230 V AC work.

1.6 APPLICATIONS OF PROPOSED SYSTEM

The proposed system is intended as a prototype and teaching platform, and the same concept can be developed into practical installations with proper engineering and certification. Possible applications are:

- **Laboratories and educational institutions:** demonstration of IoT-based condition monitoring and protection in electrical and electronics laboratories.
- **Workshops and small industries:** monitoring of the cables that feed motors, welding sets and portable machines, where voltage variation and vibration are common.
- **Temporary power connections:** construction sites, exhibitions and events where long extension cables are used under changing mechanical and environmental conditions.
- **Domestic and commercial buildings:** observation of cables feeding heavy loads such as air-conditioners, water pumps and heaters.
- **Predictive maintenance studies:** collection of cable condition data over a period of time for trend analysis and maintenance planning.

- **Remote or unattended installations:** pump houses, telecom shelters and storage rooms, where the status of a feeder cable can be checked from a mobile phone.

CHAPTER-2
LITERATURE SURVEY

2.1 LITERATURE SURVEY

The literature related to the present work is grouped under five headings: IoT-based cable monitoring, electrical cable fault detection, ESP32-based monitoring systems, sensor-based predictive maintenance and automatic electrical protection. Only published works whose bibliographic details could be identified are cited, and each summary is limited to what the cited work reports.

A. IoT-based cable monitoring

1. Design and Development of Underground Cable Fault Detection and Localization using IoT – D. G. Khairnar, R. Rathod, T. Patil, S. Madkaikar (IRJET, 2023) [1]

Key Findings:

- Detects open-circuit and short-circuit faults in an underground cable using capacitance and inductance measurement circuits.
- Uses an ATmega328 microcontroller and displays the result on an LCD and on a web page through the internet.

Challenges / Relevance to the present work:

- Concentrates on locating a fault after it has occurred rather than on continuous monitoring of voltage, temperature, vibration and magnetic field.
- Has no automatic disconnection of the supply.

2. Underground Cable Fault Detection Using IoT – P. D. Kadam, A. Nigam, V. Pandey, S. Samal (IRJET, 2020) [2]

Key Findings:

- Locates a fault in an underground cable using a NodeMCU and an AT89S52, based on the change of current or voltage with the length of cable at which the fault occurs.

- Models the cable as a set of series resistors and generates faults using switches; the fault distance is shown on a 16×2 LCD.

Challenges / Relevance to the present work:

- The system is demonstrated on a resistor model of a cable, not on a live 230 V cable.
- Does not observe environmental or mechanical conditions such as temperature and vibration.

3. Design and Detection of Underground Cable Fault Using Raspberry Pi and IoT System – Senthilkumar (IJTSRD, 2025) [3]

Key Findings:

- Proposes an underground cable fault location model using a Raspberry Pi and IoT, using the current transformer principle.
- The fault distance information is transferred through the internet.

Challenges / Relevance to the present work:

- Focuses on fault distance estimation, not on multi-sensor health monitoring.
- Uses a heavier computing platform than is needed for a simple monitoring and protection prototype.

B. Electrical cable fault detection

4. Traveling-Wave-Based Fault-Location Scheme for Multiend-Aged Underground Cable System – M. Gilany, D. K. Ibrahim, E. S. T. Eldin (IEEE Trans. Power Delivery, 2024) [4]

Key Findings:

- Presents a wavelet-based travelling wave scheme to estimate the fault location in aged cable systems.
- The scheme reduces the effect of change in propagation velocity caused by ageing of the cable, and its accuracy is studied by ATP/EMTP simulation.

Challenges / Relevance to the present work:

- Needs synchronized recorded data from the cable terminals and signal processing; it is intended for utility-level systems and not for a low-cost embedded prototype.
- Shows that cable ageing is an important concern, which supports the need for continuous condition observation.

C. ESP32-based monitoring systems

5. Comparative Analysis and Practical Implementation of the ESP32 Microcontroller Module for the Internet of Things – A. Maier, A. Sharp, Y. Vagapov (Proc. 2022 Internet Technologies and Applications, IEEE) [5]

Key Findings:

- Describes the ESP32 as a low-cost, low-power system on a chip with Wi-Fi and Bluetooth and a dual-core processor, and compares it with competing devices.
- Demonstrates a portable wireless oscilloscope with a mobile application as a practical use of the ESP32.

Challenges / Relevance to the present work:

- Confirms the suitability of the ESP32 for sensor acquisition and wireless data transfer, as used in the present work.
- The paper does not address mains-voltage electrical protection.

6. Using the ESP32 Microcontroller for Data Processing – M. Babiuch, P. Foltýnek, P. Smutný (Proc. 20th International Carpathian Control Conference, IEEE, 2021) [6]

Key Findings:

- Examines the use of the ESP32 microcontroller for data processing in embedded applications.

Challenges / Relevance to the present work:

- Supports the choice of the ESP32 as the controller for acquisition and processing of several sensor signals.
- Application-specific protection logic must still be designed and validated by the user.

7. Internet of Things is a Revolutionary Approach for Future Technology Enhancement: A Review – S. Kumar, P. Tiwari, M. Zymbler (Journal of Big Data, 2023) [7]

Key Findings:

- Reviews IoT architecture, enabling technologies and application domains.
- Discusses the challenges of deploying IoT systems.

Challenges / Relevance to the present work:

- Dependence on network connectivity and cloud services is an inherent limitation of IoT systems, so local actions such as the buzzer should not depend on the network.

D. Sensor-based predictive maintenance

8. A Review on Machinery Diagnostics and Prognostics Implementing Condition-Based Maintenance – A. K. S. Jardine, D. Lin, D. Banjevic (Mechanical Systems and Signal Processing, 2023) [8]

Key Findings:

- Describes condition-based maintenance as a process of data acquisition, data processing and maintenance decision-making.
- Discusses diagnostics (identifying a fault) and prognostics (predicting future condition).

Challenges / Relevance to the present work:

- The present prototype performs data acquisition and threshold-based decision-making; prognostics are outside its scope.

9. A Cyber-Physical Systems Architecture for Industry 4.0-Based Manufacturing Systems – J. Lee, B. Bagheri, H.-A. Kao (Manufacturing Letters, 2025) [9]

Key Findings:

- Proposes a five-level architecture (connection, conversion, cyber, cognition and configuration) for cyber-physical systems.
- Describes how sensor data can be converted into information that supports decisions and self-configuration.

Challenges / Relevance to the present work:

- The present work corresponds to the lower levels of this architecture (sensor connection and data-to-information conversion).

10. A Systematic Literature Review of Machine Learning Methods Applied to Predictive Maintenance – T. P. Carvalho et al. (Computers & Industrial Engineering, 2019) and Predictive Maintenance in the Industry 4.0: A Systematic Literature Review – T. Zonta et al. (Computers & Industrial Engineering, 2020) [10], [11]

Key Findings:

- Review the methods, data sources and application areas of predictive maintenance.
- Show that sensor data collected over time is the basis of predictive maintenance.

Challenges / Relevance to the present work:

- Machine-learning-based prediction needs large quantities of recorded data; the present prototype does not implement it and treats it as future scope.

E. Automatic electrical protection systems

11. Power System Relaying – S. H. Horowitz and A. G. Phadke (Wiley, 4th ed., 2024) [12]

Key Findings:

- Describes the principles of protective relaying, including relays that respond to abnormal voltage, and the use of relays to isolate abnormal conditions.

Challenges / Relevance to the present work:

- Utility-scale protection philosophy supports the idea of automatic isolation, but the prototype in this report is a low-cost demonstration and not a certified protective relay.

CHAPTER-3
PROJECT DESCRIPTION

3.1 PROPOSED SYSTEM DESCRIPTION

The proposed system monitors the health of a 230 V AC electrical cable using four sensors connected to an ESP32 development board. The ESP32 reads the sensors, compares the readings with the defined fault levels, drives a buzzer, and sends the readings and the overall status to the Blynk IoT cloud through Wi-Fi, from where they appear on a mobile dashboard. A 12 V relay module is provided to disconnect the Phase supply of the monitored cable when a fault occurs.

The system is divided into two electrically distinct parts. The **monitoring circuit** (ESP32, sensors and buzzer) operates from a regulated low-voltage DC supply and stays powered at all times. The **relay switching circuit** (relay contacts, Phase and Neutral wires and the monitored cable) carries the 230 V AC supply of the monitored load. The relay switches only the Phase wire; the Neutral wire bypasses the relay. As a result, the monitored cable can be disconnected without switching OFF the ESP32 and the sensors, and the system can continue to display the fault status and detect when the fault has cleared.

The parameters monitored and the fault levels defined for the project are summarised in Table 3.1. The voltage limits of 200 V and 260 V are the confirmed final protection requirement. The thresholds present in the older version of the Arduino code are different and must be replaced by the final values before the relay logic is integrated.

Table 3.1: Monitored parameters and fault levels

Parameter	Sensor / pin	Middle level	ault level (intended final)
voltage	MPT101B / GPIO 34	t defined	ow 200 V (low voltage) or ve 260 V (high voltage); mal range 200 V – 260 V
mperature	T11 / GPIO 4	°C	°C and above
gnetic field	Y-024 / GPIO 35	00 (raw ADC ding)	00 and above (raw ADC ding)
ration	V-420 / GPIO 27	t defined	gital output HIGH indicates ormal vibration
midity	T11 / GPIO 4	played only	fault level defined

3.2 OVERALL BLOCK DIAGRAM AND EXPLANATION

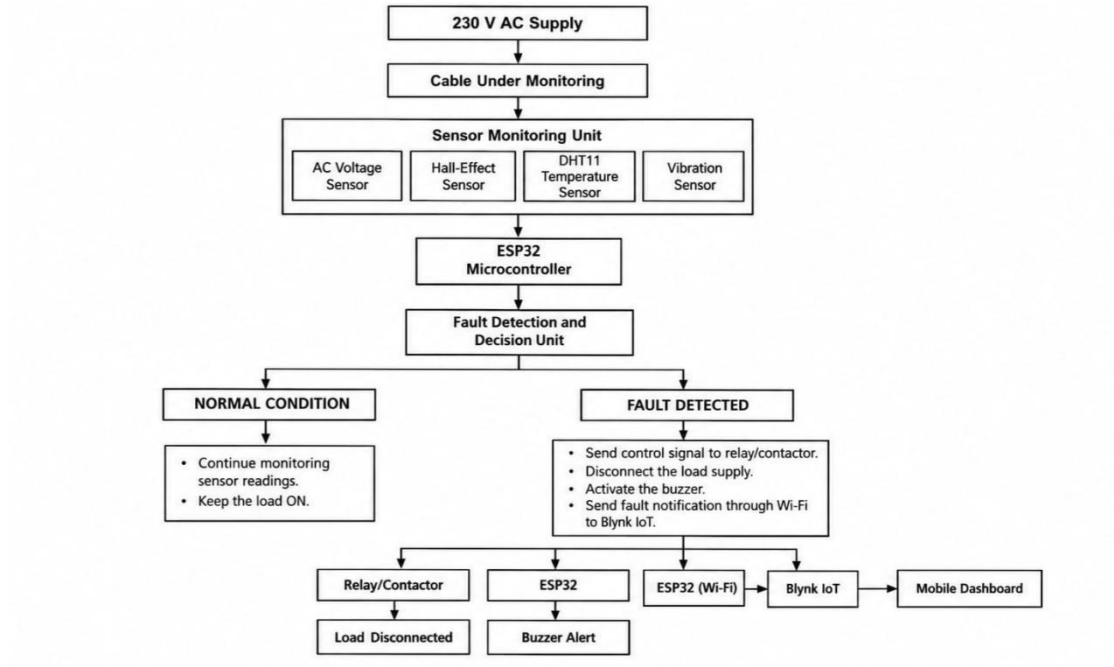


Figure 3.1: Overall block diagram of the Smart Cable Health Monitoring and Automatic Power Disconnection System

Figure 3.1 shows the functional blocks of the system. The 230 V AC mains supply feeds two paths. The first path goes to the monitoring power supply, in which a step-down transformer, bridge rectifier, filter capacitor and 7805 regulator produce

the DC supply for the ESP32 and the sensors. The second path carries the Phase wire to the COM terminal of the relay; the NO terminal provides the switched Phase to the monitored cable, and the Neutral wire goes directly to the cable.

The four sensors are connected to the ESP32 inputs. The ZMPT101B and the KY-024 give analog outputs, which are read through the ADC pins GPIO 34 and GPIO 35. The SW-420 gives a digital output read on GPIO 27, and the DHT11 sends its digital data on a single wire to GPIO 4. The ESP32 drives the buzzer through GPIO 23 and is intended to drive the relay module through GPIO 5. The ESP32 connects to a Wi-Fi router, through which the readings reach the Blynk cloud and the mobile dashboard.

3.3 CIRCUIT DIAGRAM AND EXPLANATION

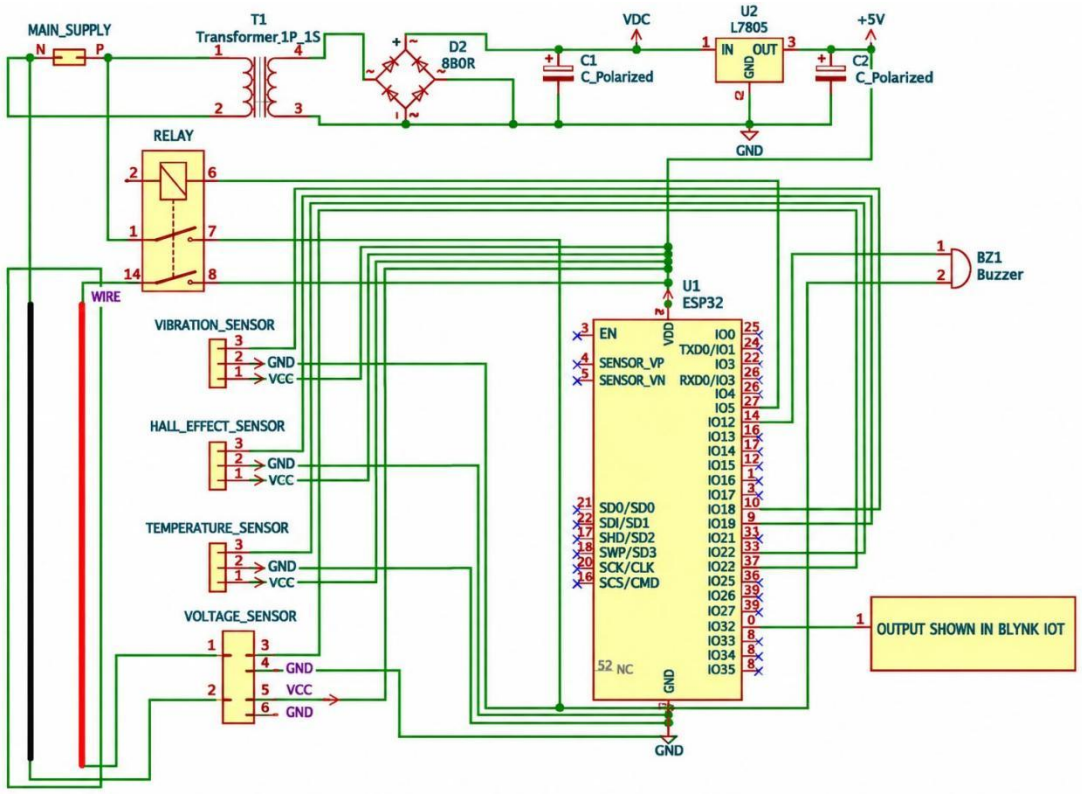


Figure 3.2: Connection diagram of the ESP32 with the sensors, buzzer and relay module

Figure 3.2 shows the signal connections between the ESP32 and the peripheral devices. The pin assignment is given in Table 3.3. The diagram is a connection diagram prepared from the project pin configuration; it shows signal connections

only. The supply connections of the individual modules follow the voltage ratings of the modules, and the supply connection of the relay module is to be confirmed from the final wiring.

Table 3.3: ESP32 pin configuration

Device	ESP32 pin	Function
ltage sensor	GPIO 34	alog input – supply voltage asurement
ll effect sensor	GPIO 35	alog input – magnetic field reading
ration sensor	GPIO 27	gital input – HIGH indicates ormal vibration
mperature sensor	GPIO 4	gle-wire digital data – temperature l humidity
zzer	GPIO 23	gital output – alarm indication
ay module (control ut)	GPIO 5	gital output – planned connection; y logic is not present in the shared luino sketch

GPIO 34 and GPIO 35 are input-only pins of the ESP32 and belong to the first ADC unit (ADC1). The second ADC unit (ADC2) cannot be used while Wi-Fi is active, and the selection of ADC1 pins for the two analog sensors is therefore appropriate for a Wi-Fi-connected application. The ESP32 ADC is a 12-bit converter, so a raw reading lies between 0 and 4095; the Hall sensor levels of 2000 and 3000 are raw readings on this scale.

3.4 POWER SUPPLY DESIGN

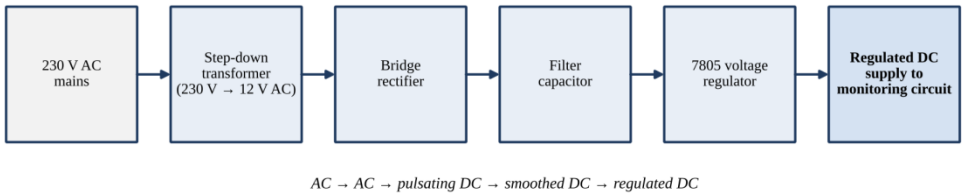


Figure 3.3: Power supply chain of the monitoring circuit

The monitoring circuit is powered from a linear DC supply that is derived from the 230 V AC mains through the chain shown in Figure 3.3: 230 V AC → step-down transformer → 12 V AC → bridge rectifier → filter capacitor → 7805 voltage regulator → DC supply. This supply is taken from the un-switched mains, ahead of the relay, so that it remains available when the relay disconnects the monitored cable.

3.4.1 Function of each stage

- **Step-down transformer:** reduces the 230 V AC mains to 12 V AC and provides galvanic isolation between the mains and the low-voltage electronics.
- **Bridge rectifier:** four diodes arranged as a bridge convert the alternating voltage into a pulsating DC voltage with both half cycles utilised (full-wave rectification).
- **Filter capacitor:** charges near the peaks of the rectified waveform and supplies the load between peaks, which reduces the ripple of the pulsating DC.
- **7805 voltage regulator:** a fixed three-terminal regulator that produces a regulated +5 V output from the unregulated input, provided that the input voltage stays sufficiently above the output (about 7 V or more) and the load current stays within the rating of the device.

3.5 HARDWARE COMPONENTS USED

Table 3.4: Hardware components used in the project

.No	Component	Role in the project	uantity
1	P32 development rd	in controller – sensor uisition, fault evaluation, buzzer l relay control, Wi-Fi nmunication	1
2	IPT101B AC voltage sor	asures the AC supply voltage of monitored cable	1
3	-024 Hall effect sor	ses the magnetic field near the le	1
4	-420 vibration sor	ects abnormal vibration	1
5	T11 sensor	asures temperature and humidity	1
6	V relay module	itches the Phase supply of the nitored cable	1
7	zzer	dible alarm during a fault	1
8	p-down transformer	V AC to 12 V AC for the nitoring supply	1
9	dge rectifier	l-wave rectification of the 12 V	1
10	er capacitor	oothing of the rectified voltage	1
11	05 voltage regulator	gulated DC supply	1

3.6 SOFTWARE USED

3.6.1 Arduino IDE and Embedded C/C++

The firmware of the ESP32 is written in Embedded C/C++ using the Arduino IDE with the ESP32 board support package. The program follows the usual two-function structure of an Arduino sketch: an initialisation routine that configures the

pins, the serial port, the sensor and the Wi-Fi/Blynk connection, and a main loop that repeats the sensing, evaluation and reporting cycle. The sketch uses the Blynk library for the ESP32 and a DHT sensor library.

3.6.2 Blynk IoT

Blynk IoT is a platform that connects hardware to a cloud server and provides mobile and web dashboards [17]. The ESP32 sends data to numbered virtual pins (V0 to V5), and the dashboard widgets display the values of these datastreams. Section 3.9 describes the project dashboard.

3.6.3 Wi-Fi communication

The ESP32 connects to a Wi-Fi access point and communicates with the Blynk cloud over the internet. The local buzzer indication does not depend on the network; the dashboard, however, can be updated only when the Wi-Fi network and the internet are available.

3.6.4 Status of the software

Reading the four sensors, evaluating the status and sending the values to the Blynk dashboard are part of the shared sketch. **The shared sketch does not contain relay control logic**, even though the relay hardware has been tested separately. The voltage thresholds in the earlier code are also different from the final protection requirement. In addition, the present ZMPT101B calculation is approximate and has not been confirmed as a calibrated RMS measurement. These three points are carried to the limitations in Chapter 5.

3.7 SENSOR WORKING PRINCIPLES

3.7.1 ZMPT101B AC voltage sensor

The ZMPT101B contains a small voltage transformer that is used as a current transformer in the sense that a resistor in series with the mains side limits the primary current to a few milliamperes. The secondary current is converted to a

small voltage and conditioned by an operational amplifier stage on the module. The output is an AC waveform at the mains frequency that is centred on a DC offset, with an amplitude proportional to the mains voltage. Because it uses a transformer, the module provides galvanic isolation between the mains input and the output pin.

The ESP32 samples this waveform with its ADC. A true RMS value is obtained by sampling many points over one or more complete cycles, subtracting the DC offset from every sample, squaring, averaging and taking the square root, and then multiplying by a calibration constant: $V(rms) = k \times \sqrt{[(1/N)\ \Sigma\ (v(i) - v(offset))^2]}$. The constant k is found by comparing the module reading with a reference meter at one or more voltages. The present calculation in the sketch is approximate and has **not** been confirmed as a calibrated RMS measurement, so the voltage values should be treated as indicative until this calibration is completed.

3.7.2 KY-024 Hall effect sensor

A Hall element produces a voltage proportional to the magnetic flux density that passes through it, when it is supplied with a bias current (Hall effect). In a linear Hall module the analog output sits at a quiescent level when no magnetic field is present and moves up or down depending on the polarity and strength of the field. A straight conductor carrying a current I produces a magnetic flux density $B = \mu 0\ I / (2\pi\ r)$ at a distance r, so a change in the current or in the position of the cable changes the Hall output [20].

The ESP32 reads the analog output as a raw number between 0 and 4095. For this project, 2000 is the middle level and 3000 and above is the fault level. These are relative raw levels that were chosen for the prototype; the readings are not calibrated in gauss or amperes. In a two-core cable the Phase and Neutral currents are equal and opposite, so the external magnetic field is small; the sensor position and the exposure of the Phase conductor affect the reading, and this must be considered when the sensor is installed.

3.7.3 SW-420 vibration sensor

The SW-420 module uses a vibration-sensitive switch element whose contact condition changes when the module is shaken or vibrated. An LM393-type comparator on the module compares the signal with a threshold set by a trimmer and gives a digital output. In this project a HIGH level on GPIO 27 indicates abnormal vibration. The trimmer must be adjusted so that normal handling does not give false indications while genuine vibration is detected.

3.7.4 DHT11 temperature and humidity sensor

The DHT11 contains a resistive element whose resistance changes with temperature and a capacitive element whose capacitance changes with humidity, together with a small controller that converts the signals into a digital word. The data is sent to the ESP32 over one wire by a timed pulse protocol. A new reading can be requested about once every second. The temperature middle level is 30 °C and the fault level is 35 °C and above. The DHT11 senses the air temperature at its own location, so it should be placed close to the cable if it is to represent the cable temperature, and the reading is not the temperature of the conductor itself.

3.7.5 Buzzer

The buzzer connected to GPIO 23 gives a local audible alarm. Its activation does not require the network connection, so the alarm still works when Wi-Fi is unavailable. Whether the buzzer is an active or a passive type is Interconnected.

3.8 RELAY-BASED AUTOMATIC POWER DISCONNECTION

3.8.1 Relay wiring

The Phase wire of the mains is connected to the COM terminal of the relay, and the NO terminal provides the switched Phase to the monitored cable. The Neutral wire does not pass through the relay and goes directly to the cable. When the relay coil is energised, COM is connected to NO and the cable receives the supply. When the coil is de-energised, the NO contact opens and the Phase supply to the cable is removed.

Because the monitored cable is supplied through the NO contact, a loss of the control signal or of the relay supply also disconnects the cable. The relay therefore opens in the safe direction when the monitoring system itself fails. The polarity of the control signal needed to energise the relay (active-HIGH or active-LOW) depends on the module and must be confirmed during integration .

3.8.2 Monitoring circuit and relay switching circuit

Table 3.6: Difference between the monitoring circuit and the relay switching circuit

Aspect	Monitoring circuit	Relay switching circuit
rpose	ıses the condition of the ble, raises the alarm, reports Blynk	nnects or disconnects the se supply of the monitored ble
in parts	P32, ZMPT101B, KY-024, -420, DHT11, buzzer	ay contacts (COM, NO), se and Neutral wires, nitored cable
ltage level	w-voltage DC from the 05-based supply (sensor ins input side of the IPT101B is an exception)	) V AC
wer source	nsformer, rectifier, acitor and regulator chain en from un-switched mains	ins Phase through relay M and NO
naviour during a lt	ntinues to operate and nitor	se supply of the monitored le is disconnected ended)

3.8.3 Intended disconnection and reconnection logic

The intended logic is as follows. If any one of the monitored parameters enters its fault range, the ESP32 switches the buzzer ON, de-energises the relay so that the Phase supply of the monitored cable is disconnected, and sets the overall status on the dashboard to FAULT. As the ESP32 and the sensors are still powered, the

readings continue to be updated. When all the parameters return to acceptable values, the buzzer is switched OFF, the relay is energised again, and the supply to the cable is restored. This logic is the design intention. It is not part of the shared Arduino sketch.

Two design points need attention when this logic is implemented. First, if the ZMPT101B measures the voltage on the switched side of the relay, the measured voltage becomes zero after disconnection and the low-voltage fault would never clear. For automatic reconnection based on voltage recovery, the sensor must measure the un-switched (source) side. Second, a fault that is caused by the load current itself (for example, a high Hall reading) may disappear as soon as the cable is disconnected, which would cause the relay to switch ON and OFF repeatedly. A delay before reconnection or a hysteresis band is therefore recommended.

3.8.5 Electrical safety limitations for 230 V AC implementation

The prototype works with the 230 V AC mains, which can cause electric shock, burns and fire. The following precautions and limitations apply:

- The mains connections, the relay terminals, the transformer primary and the ZMPT101B input must be fully enclosed in an insulated enclosure with no exposed live terminals, with proper cable glands and strain relief.
- The prototype must be supplied through a suitably rated MCB or fuse and, where available, an RCCB, and the metallic parts must be earthed. The relay disconnection is not a substitute for these devices.
- Only the Phase wire is switched. When the relay is open, the Neutral wire (and the supply side of the relay) remains connected, and the cable must **not** be considered electrically isolated or safe to touch. For maintenance, the supply must be isolated upstream and checked with a tester.
- The Phase and Neutral must be connected with the correct polarity. If the polarity of the mains socket is reversed, the relay will switch the Neutral instead of the Phase.

• The relay contacts must be rated above the maximum load current and voltage, and the load used in the prototype should be small.

• Creepage and clearance between the mains tracks and the low-voltage tracks must be maintained, and the 230 V wiring should be separated from the ESP32 and the sensor wiring.

• The prototype has not been tested to any electrical safety or EMC standard and is intended for demonstration under the supervision of qualified faculty or laboratory staff. Installations in buildings must follow the applicable wiring standards (for example, the IEC 60364 series for low-voltage installations [19] and the corresponding Indian standards).

3.9 BLYNK IOT IMPLEMENTATION

The Blynk dashboard shows the readings of the sensors and the overall status of the cable. The ESP32 sends the values to the Blynk cloud, and the dashboard widgets are linked to the virtual pins listed in Table 3.7.

Table 3.7: Blynk virtual pin (datastream) assignment

irtual pin	Data displayed	Source
V0	ltage	IPT101B (GPIO 34)
V1	ll sensor value	-024 (GPIO 35)
V2	mperature	T11 (GPIO 4)
V3	midity	T11 (GPIO 4)
V4	ɔration status	-420 (GPIO 27)
V5	erall status: NORMAL / DDLE LEVEL / FAULT	aluated by the ESP32 from the sor values

The present dashboard displays the fault status. Mobile push notifications are **not** implemented in the prototype. Blynk offers an event and notification feature that could be used in future to alert the user on the phone when the status changes to FAULT.

3.10 COMPLETE WORKING PRINCIPLE

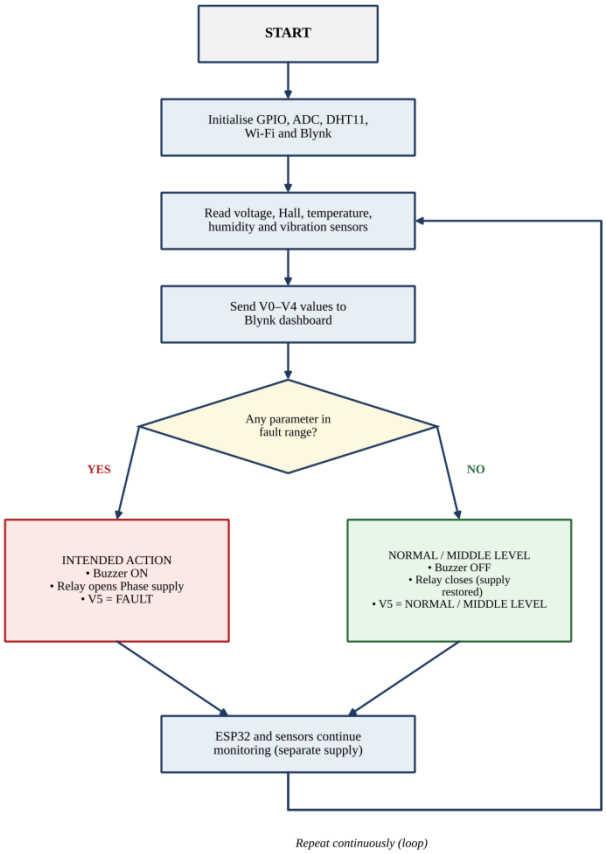


Figure 3.5: Flowchart of the intended operation of the system (relay branch is the design intention)

When the system is powered, the 230 V AC mains energises the monitoring supply, and the ESP32 starts and initialises its pins, the DHT11, the Wi-Fi connection and the Blynk connection. The ESP32 then repeatedly reads the voltage, Hall, temperature, humidity and vibration values and sends them to the corresponding Blynk virtual pins.

In each cycle the readings are compared with the fault levels in Table 3.1. If every parameter is in the normal range, the status is NORMAL; if a parameter is around its middle level, the status is MIDDLE LEVEL; if a parameter is in its fault range, the status is FAULT. In the intended operation shown in Figure 3.5, a FAULT status activates the buzzer and de-energises the relay so that the Phase supply of the monitored cable is disconnected. The ESP32 and the sensors continue to be

powered by the separate monitoring supply, so the readings and the FAULT status remain visible on the dashboard. When the fault clears, the buzzer is switched OFF and the relay is energised again to restore the supply.

Table 3.8: Intended system response for each condition

Condition	**Buzzer**	**hase supply of onitored cable**	**Blynk status (V5)**
parameters normal	OFF	Connected	NORMAL
ameter near middle el	OFF	Connected	MIDDLE LEVEL
y parameter in fault ge	ON	Disconnected (intended)	FAULT
ılt cleared	OFF	Reconnected automatically (intended)	NORMAL

CHAPTER-4
RESULTS AND DISCUSSION

4.1 HARDWARE PROTOTYPE AND EXPLANATION

The prototype integrates the ESP32 development board, the four sensors, the buzzer, the relay module and the monitoring power supply on a common assembly. The monitoring circuit and the relay switching circuit are kept separate, and the 230 V AC connections are confined to the transformer primary, the ZMPT101B mains input and the relay contacts. Figure 4.1 shows the assembled prototype and Figure 4.2 shows the arrangement of the sensors near the monitored cable.

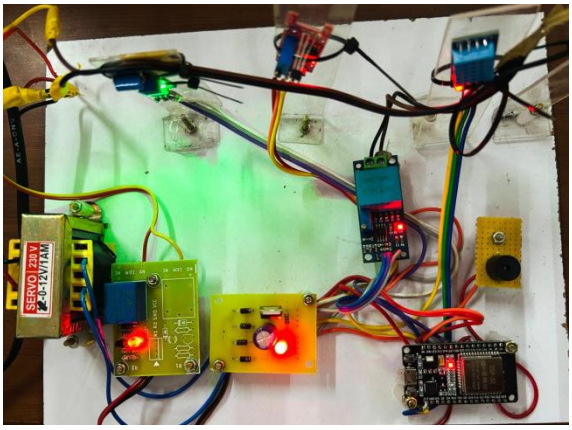


Figure 4.1: Hardware prototype of the Smart Cable Health Monitoring and Automatic Power Disconnection System

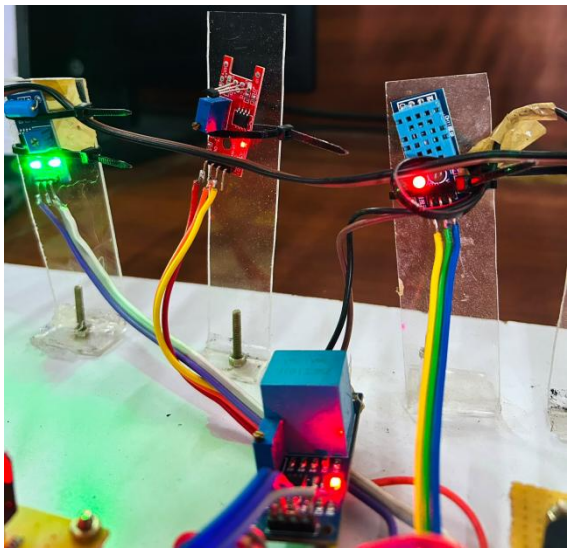


Figure 4.2: Placement of the sensors near the monitored cable

4.8 BLYNK DASHBOARD RESULTS

The Blynk dashboard receives the voltage (V0), Hall sensor value (V1), temperature (V2), humidity (V3), vibration status (V4) and overall status (V5). The status text can be NORMAL, MIDDLE LEVEL or FAULT. The dashboard displays the fault status, but mobile push notfications are not implemented.

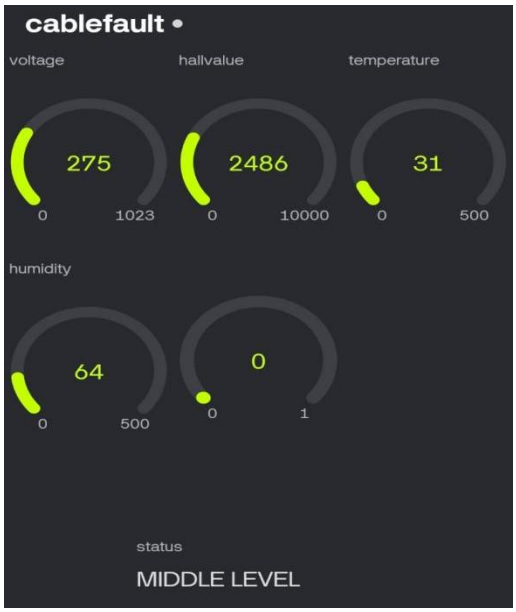


Figure 4.8: Blynk dashboard showing normal operation

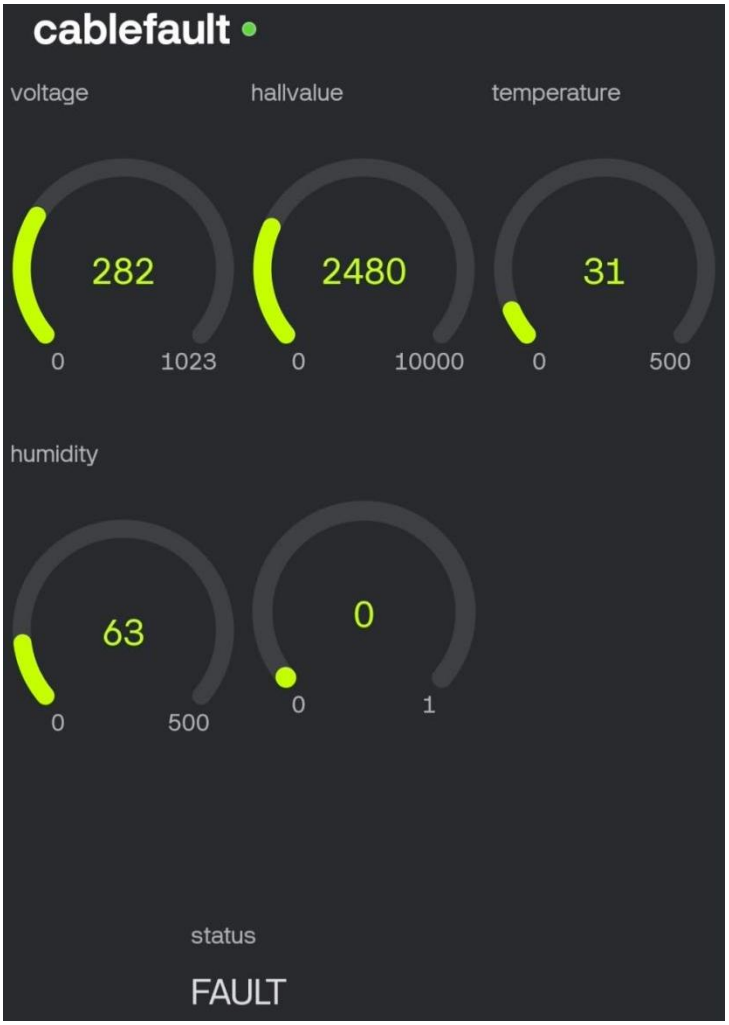


Figure 4.9: Blynk dashboard showing fault status

4.9 OVERALL SYSTEM PERFORMANCE

The prototype demonstrates that the four parameters of a cable can be sensed by an ESP32, that the readings can be viewed on a mobile phone, and that the relay hardware can switch the Phase of a cable while the monitoring circuit stays on a separate supply. Table 4.5 states the position of each function, so that the reader can see which functions are completed and which are pending.

Table 4.5: Status of the functions of the prototype

Function	Status	Remark
ltage sensing MPT101B)	sted individually	culation approximate; IS calibration not nfirmed
mperature and humidity sing (DHT11)	sted individually	ading represents air mperature at the sensor
ɔration detection (SW-420)	sted individually	ɔends on sensitivity setting
gnetic field sensing (KY-4)	sted individually	w, uncalibrated readings
zzer alarm	ended on fault	nfirm in final test
ay hardware switching of se	sted (hardware)	M = Phase in; NO = tched Phase; Neutral ɔasses
ay control in software	t integrated	t present in the shared tch
tomatic reconnection after lt clears	t demonstrated	ɔends on relay logic and sor placement
nk dashboard (V0–V5)	ɔlemented	lt status shown; no push ification
ɔarate monitoring power ɔply	signed and embled	asured values

CHAPTER-5
CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

A Smart Cable Health Monitoring and Automatic Power Disconnection System was designed and built as a prototype using an ESP32, a ZMPT101B voltage sensor, a KY-024 Hall effect sensor, an SW-420 vibration sensor, a DHT11 sensor, a buzzer and a 12 V relay module. The monitoring circuit is powered from a separate regulated supply made of a step-down transformer, bridge rectifier, filter capacitor and 7805 regulator, so that it is not affected when the monitored cable is disconnected.

The voltage, temperature, vibration and Hall sensor variations were tested individually, and the relay hardware was tested separately. The readings and the overall status (NORMAL / MIDDLE LEVEL / FAULT) are displayed on a Blynk IoT mobile dashboard using the virtual pins V0 to V5. The relay wiring, in which COM receives the Phase, NO provides the switched Phase and the Neutral bypasses the relay, allows the monitored cable to be disconnected without switching OFF the monitoring system.

The project also identified the work that is yet to be completed: integration of the relay control logic in the sketch, update of the voltage thresholds to the final 200 V and 260 V limits, calibration of the ZMPT101B measurement, and implementation of push notifications. The system is a demonstration prototype and not a certified protection device.

5.2 LIMITATIONS

- The shared Arduino sketch does not contain the relay control logic; the relay hardware has been tested, but automatic disconnection and reconnection by the sketch have not been demonstrated.

- The thresholds in the older code differ from the intended final voltage protection limits (below 200 V and above 260 V).
- The ZMPT101B calculation is approximate and has not been confirmed as a calibrated RMS measurement.
- The Hall sensor readings are raw, uncalibrated values; in a two-core cable the external field is small because the Phase and Neutral currents cancel.
- The DHT11 has limited resolution and sampling rate and measures the air temperature at the sensor, not the conductor temperature.
- The dashboard displays the fault status, but mobile push notifications are not implemented, and the dashboard depends on Wi-Fi and internet availability.
- Only the Phase wire is disconnected, so the cable must not be treated as isolated; the prototype does not replace MCBs, RCCBs or fuses, and has not been tested to electrical safety or EMC standards.
- Without a reconnection delay or hysteresis, a fault caused by the load current may cause repeated switching of the relay.
- Numerical performance figures (accuracy, response time and detection rate) have not been measured and are not reported.

5.3 FUTURE SCOPE

- Integrating the relay control logic with the fault evaluation, with a reconnection delay or hysteresis, and testing the complete protection sequence.
- Updating the thresholds to the final limits and calibrating the ZMPT101B using true RMS sampling against a reference meter.
- Adding a current sensor so that load current and overload conditions are measured directly.
- Adding Blynk events and mobile push notifications, and local data logging so that readings are retained when Wi-Fi is not available.
- Storing the sensor history on a cloud platform for trend analysis and predictive maintenance, including simple trend-based or machine-learning-based prediction.

- Using a thermistor or thermocouple fixed on the cable surface in place of the DHT11 for a more direct measurement of cable temperature.
- Designing a printed circuit board with proper isolation, an enclosure, fuse protection and an isolating relay driver to a proper safety standard.
- Extending the system to multiple cables with one controller and adding over-the-air updates of the firmware.

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